

BrainBook README

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Access & Setup

The Prototype can be accessed [here](#). No additional setup needs to be done once the prototype is loaded; simply begin clicking the prototype to start!

- The Prototype should begin on the "Closed Page" (See Image (1) Below). If it does not, navigate back by clicking various 'X' buttons.

The prototype should *only* be run on an **iPad Pro 12.9"**. This prototype attempts to replicate the flow of our hypothetical physical device (with multiple adjacent screens) onto a single upright screen. This translates well to the Settings Page, Existing Questions, and Initial Page, but it does not work as well for the Main Interaction Page. In our hypothetical device, this page consists of two adjacent screens, which cannot easily be replicated in Figma.

Our solution was to make the Main Interaction Page consist of two adjacent, clearly-delimited pages. The user can scroll between these two pages on the iPad screen manually to see both the AI and User space.

Context

BrainBook is an educational device that challenges the user to reason about a question by spoonfeeding them snippets of a full explanation and asking them expansive questions. The product mostly revolves around the "learning loop," where the AI and the user go back and forth to guide the user towards realizing the answer themselves.

The user should be able to:

1. Begin a new interaction by completing the initial three prompts (Keywords, Question, and Previous Knowledge) and opening the product
 - a. The User may also be able to continue existing questions
2. Change their settings and set preferences such as responding with text or diagrams, and favorite tools

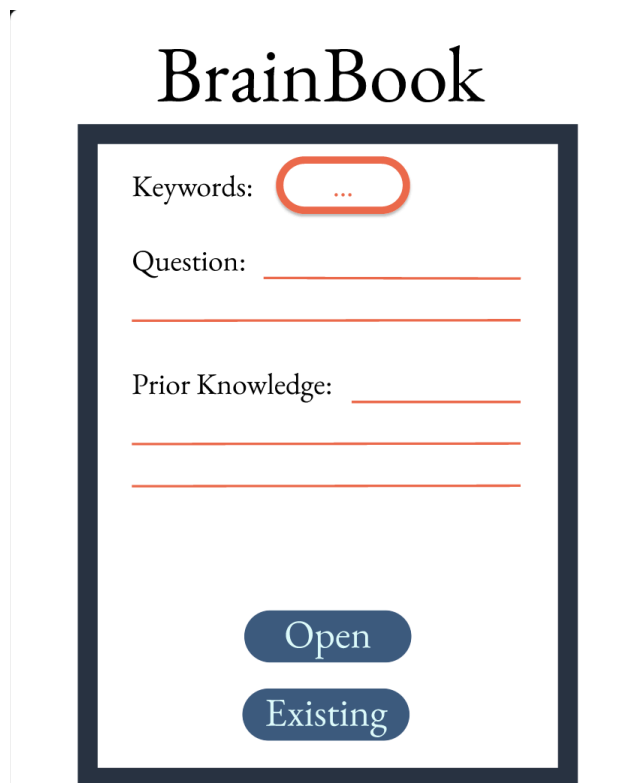
3. Interact with the AI to initiate the core learning loop by responding to the initial AI prompt and getting feedback from the AI that then prompts the user to respond again

After the user finishes using the product, they should have a deep understanding of their specific question that they discovered themselves under the guidance of the product.

Instructions

In the hypothetical product, users would directly interact with the product by writing their prompts or responses. Figma does not support writing, so prompts and responses are *pre-loaded* onto the prototype, and generally accessed with a click.

Buttons in the final product are colored normally and may be clicked normally. These buttons include icon buttons (like the Settings dial or the 'X' button) and text buttons (like the "Open" or "Return" button). Prompts that are meant to be filled in with writing are highlighted in **orange** and may be clicked to fill in their content with the pre-loaded response.



The image shows a mockup of a mobile application interface titled "BrainBook". The interface is contained within a dark blue rounded rectangle. At the top, the title "BrainBook" is displayed in a large, black, serif font. Below the title, there are three input fields, each with a label and a red outline. The first field is labeled "Keywords:" and contains three dots "...". The second field is labeled "Question:" and has a single red line. The third field is labeled "Prior Knowledge:" and has two red lines. At the bottom of the form, there are two blue buttons with white text. The first button is labeled "Open" and the second button is labeled "Existing".

(1) The orange elements may be clicked to be filled in

BrainBook

Keywords:

Question:

Prior Knowledge:

(2) The blue-shaded elements may be clicked. Notice how the text has been filled in following user clicks

To achieve simple task flow #1 (Begin a new interaction):

1. Click the Orange pill next to the "Keywords" text
 - This will fill the pill in with text – a keyword related to the user's question
2. Click the Orange lines next to the "Question" text
 - This will fill those lines in with text – the user's question
3. Click the Orange lines next to the "Prior Knowledge" text
 - This will fill those lines in with text – the user's prior knowledge
4. Click the "Open" Button once the three prompts have been completed
 - This will open the main page
 - Clicking this button prematurely will open a Warning screen

Settings



Preferred Learning Style:



Favorite Learning Tools:

- Games _____
- Free Responses _____
- _____
- _____

(3) The Settings Page has Preferred Learning Styles (with the style circled) and Favorite Learning Tools – in this screenshot, some learning tools have been filled in by clicking the orange lines

To achieve moderate task flow #2 (Change the settings):

1. Once the main page is open, click the Settings icon next to the "AI Space" text (the gear icon)
 - This will open the Settings Page
 - This page contains two main modifiable preferences:
 - Preferred Learning Style
 - Favorite Learning Tools
2. To change the preferred Learning Style, click the desired option ("Text" or "Diagrams" within the Orange pills)
 - The currently selected Learning Style is circled in orange
 - This design aims to replicate how the actual product will work, where the user will actually circle their preferred style
3. To add favorite learning Tools, click on each Orange line underneath the "Favorite Learning Tools" text
 - Each orange line corresponds to a specific tool
 - Click as many or as few (up to 4) as is desired
4. To exit the Settings page, click the 'X' icon in the top right corner
 - This will navigate back to the main page

AI Space



Integration is the inverse operation of differentiation. It reconstructs a function from its derivative.

Let's start with the power rule. What's the power rule for derivatives?

Right! So what if we wanted to go the other way? For example, let's take $y = x^2$. What's the derivative of that using the formula?

Now, let's go from $2x$ to x^2 . What do you have to do to $2x$ to get to x^2 ?

Correct. Let's formulate that. We say that $(2x) * x/2$ gets us to x^2 . Now, what if we used n instead of 2 ?

Well, in that case, the formula is the exact same! If we have nx , we divide by n and multiply x : $(nx) * x/n$. Algebraically, that is just equal to x^2 . But what if we just use x ? Can you simplify?

(4) The AI Space is continually filled in on clicks. Notice the Settings dial to open the Settings page.



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It should be $nx^{(n-1)}$.

The derivative is $2x$, cus its $(2)x^{(2-1)}$ =

Okay, well we need to divide by 2 and r

Oh interesting. It would be x^2/n .

(5) To view the User Space, the user can horizontally scroll. Here, the user is halfway between the AI and the User Space.



User Space

It should be $nx^{(n-1)}$.

The derivative is $2x$, cus its $(2)x^{(2-1)} = 2x$

Okay, well we need to divide by 2 and multiply by x .

Oh interesting. It would be $x^{2/n}$.

(6) The User Space is continually filled on click. Notice the 'X' button to close the Main Page.

To achieve complex task flow #3 (interacting with the AI to begin the learning loop):

1. Once on the main page, click anywhere on the screen to "progress" the interaction
 - This page consists of two screens (the AI and User Space) – scroll horizontally to see either the AI or User space
 - The AI has already prompted the user so that the first click will show a User response
 - The User and AI go back and forth, so each click alternates between showing a User or AI response
2. To best simulate the interaction, read the AI response, click, read the User response, click, and repeat this loop
 - Each response from the AI or User builds on or responds to the other – just like a real interaction
 - Eventually, the AI space will go to the end of the page; progressing the interaction will clear both the AI space and User space, and show the next AI response
 - Progressing the interaction will work as previously described
3. Eventually, the user will say "I understand" and thank the AI
 - The interaction has *completed*, and clicking will not progress the interaction any further
 - To exit back to the initial page, click the 'X' button next to the "User Space" text

- This will return to an unfilled initial page
- Clicking the prompts will fill them in with the same information as prior, and opening the book will return to the previous, completed interaction
- This is a limitation of Figma, since we can only hardcode so many questions and interactions

Once an interaction has been initiated, the user can exit back to the initial page at any time by clicking the 'X' button next to the "User Space" text. The interaction can be accessed again by clicking the "Existing" button, which will open a page with existing interactions. Click the Right Arrow icon to open the interaction.

Limitations

Figma heavily limits the prototype in comparison to the hypothetical product:

- Because only one frame can be used for the entire prototype (an upright iPad), the design of the main page is limited to requiring horizontal scrolling to show two adjacent screens
 - The hypothetical product is intended to be a foldable, which means that one iPad screen size is intended to open up into two, adjacent iPad screens (for the main page)
 - That was implemented in Figma with horizontal scrolling, so the adjacent screens "overflow" the iPad frame
- Figma does not allow user text entry, so user writing is replicated with hard-coded responses that can be filled in by clicking orange elements
 - This also means that only one hardcoded prompt can be used for the prototype
 - This also means that the options on the Settings pages must be hardcoded
- Because user prompts are hardcoded, AI responses must also be hardcoded to respond precisely
 - Additionally, because it isn't possible to actually feed user responses into an LLM in Figma, the AI responses must be pre-determined
 - The user cannot actually interact with an AI
- Because the user can only click (and not circle or draw), the "Preferred Learning Style" buttons must be clicked, not circled

Hard-Coded Items

Due to the limitations described above, all of these items are hardcoded to simulate the product's functionality in the prototype:

- All Initial User prompts (Keywords, Question, Prior Knowledge)
- All AI Responses in the AI Space
- All User Responses in the User Space
- All options for the "Favorite Learning Tools" on the Settings Page
- All existing interactions on the Existing Interactions Page